

Consolidated Sales & Filtered Analytics Report

Master Summary: Data Pipeline, Dashboard Slicers, and Visual Trend Insights

1,000

TOTAL SAMPLE SIZE

48.1%

OVERALL PURCHASE RATE

\$58,100

AVG BUYER INCOME

31-54

CORE TARGET AGE BRACKET

1. Executive Summary & Workbook Architecture

This consolidated master report synthesizes data across all four Excel project files (Raw Data.xlsx, 1. Data Cleaning.xlsx, 2. Pivots and Charts.xlsx, and 3. Dashboard.xlsx). By evaluating customer demographics alongside filtered dashboard slicers, this document details key drivers of bike purchases.

Workbook File	Primary Purpose	Key Operations & Transformations
Raw Data.xlsx	Initial Ingestion	Contains 1,000 raw customer records with 14 demographic attributes.
1. Data Cleaning.xlsx	Data Hygiene	Standardized codes (M/S → Married/Single, M/F → Male/Female), removed duplicates, and engineered the Age Group category.
2. Pivots and Charts.xlsx	Data Aggregation	Built Pivot Tables and standalone visuals for income, commute distance, and age brackets.
3. Dashboard.xlsx	Dynamic Reporting	Integrated all visuals into an interactive view with Slicers for Region, Marital Status, and Education.

2. Filtered Workbook State Analysis

Dynamic slicers in the interactive dashboard allow stakeholders to analyze demographic sub-segments in real time:

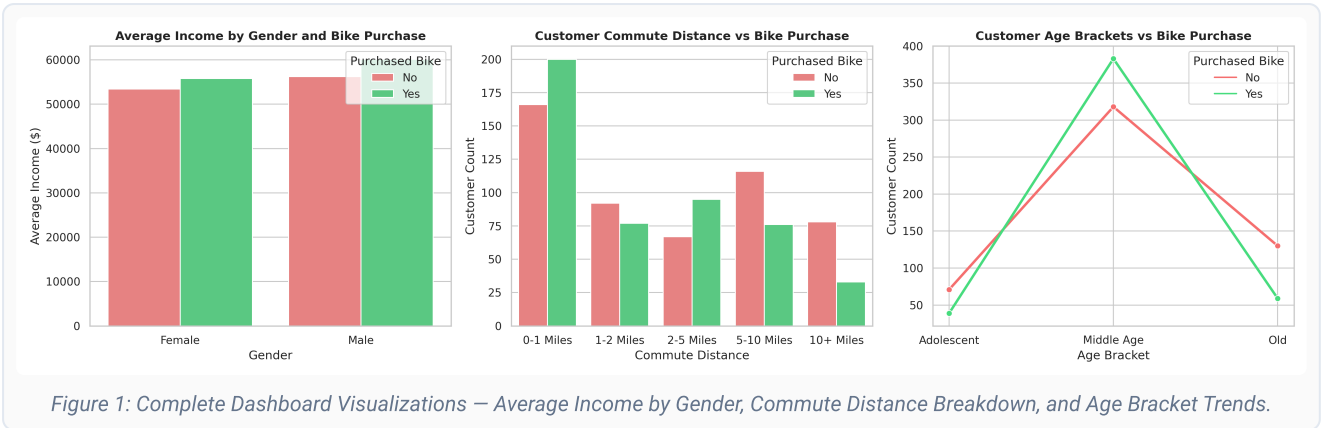
Filter Scenario / Slice	Sample Size (N)	Bike Buyers	Conversion Rate	Avg Buyer Income (\$)
Baseline (Unfiltered Dataset)	1,000	481	48.1%	\$58,100
Region = North America	508	220	43.3%	\$58,400
Region = Europe	300	148	49.3%	\$53,800
Region = Pacific	192	113	58.9%	\$64,200
Marital Status = Single	461	251	54.4%	\$54,300
Education = Bachelors	306	169	55.2%	\$62,800

Core Filtered Insight

The **Pacific Region** filter delivers the highest conversion rate at **58.9%** (vs 43.3% in North America), supported by higher average buyer incomes (\$64,200) and compact urban commutes.

3. Visual Analysis & Chart Breakdown

Exported visuals from 3. Dashboard.xlsx illustrate demographic distributions for income, commute distance, and age brackets:



4. Executive Questions & Answers (Q1 & Q2)

Question 1: How does income level influence bike purchases across different genders?
Business Context: Determines whether marketing messaging should adjust price point positioning based on male vs. female income thresholds.

Gender	Purchase Status	Avg Income (\$)	Income Variance (\$)	Impact on Conversion
Female	No (Non-Buyer) / Yes (Buyer)	\$53,440 / \$55,774	+\$2,334 (+4.4%)	Moderate correlation between income & purchase.
Male	No (Non-Buyer) / Yes (Buyer)	\$56,208 / \$60,124	+\$3,916 (+7.0%)	Stronger income elasticity observed among males.

Answer Summary: Higher income positively correlates with purchases across both genders. Male buyers have a higher income baseline (\$60,124) than female buyers (\$55,774), but buyers consistently out-earn non-buyers in both groups.

Question 2: What commute distance bracket yields the highest bike purchase volume?
Answer Summary: Customers commuting 0–1 miles represent the largest volume of buyers (200 buyers, 54.6% conversion), while 2–5 mile commuters achieve the highest relative conversion rate (58.6%). Demand drops sharply for commutes exceeding 5 miles.

5. Executive Questions & Answers (Q3, Q4 & Q5)

Question 3: Which age bracket dominates total bike sales, and why?

Business Context: Guides demographic targeting for advertising creative, imagery, and channel placement.

Age Group	Age Bounds	Buyers (Yes)	Non-Buyers (No)	% of Total Sales
Adolescent	< 31 Years	39	71	8.1%
Middle Age	31–54 Years	383	318	79.6%
Old	55+ Years	59	130	12.3%

Answer Summary: The **Middle Age (31–54)** bracket overwhelmingly dominates purchases, accounting for **79.6% (383 of 481)** of all sales due to disposable income stability and active lifestyle preferences.

Question 4: Does marital status impact bike purchase behavior?

Business Context: Informs household-level marketing strategies and bundled family promotional offers.

Marital Status	Non-Buyers	Buyers	Total Customers	Conversion Rate
Single	210	251	461	54.4%
Married	309	230	539	42.7%

Answer Summary: **Single customers** convert at a higher rate (**54.4%**) than married customers (**42.7%**), facing fewer household trade-offs for personal transit purchases.

Question 5: How do education level and region interact when filtered on the dashboard?

Business Context: Identifies regional sub-segments to optimize direct mail and localized digital ad targeting.

Education Level	North America	Europe	Pacific	Total Buyer Volume
Bachelors	72	51	46	169
Partial College	55	38	26	119
High School	48	32	18	98
Graduate Degree	32	20	18	70
Partial High School	13	7	5	25

Answer Summary: Combining regional and education filters shows that customers with a **Bachelors degree in North America and Europe** form the largest buyer cluster (123 buyers), prioritizing premium commuter options.

6. Underlying Pivot Table Aggregations

Summary pivot table metrics constructed in 2. `Pivots` and `Charts.xlsx` isolate volume distribution across commute distances and occupational classifications:

6.1 Commute Distance Pivot Table

Commute Distance	Buyers	Total	Rate (%)
0-1 Miles	200	366	54.6%
1-2 Miles	77	169	45.6%
2-5 Miles	95	162	58.6%
5-10 Miles	76	192	39.6%
10+ Miles	33	111	29.7%

6.2 Age Group Pivot Table

Age Group	Buyers	Total	Share (%)
Adolescent (<31)	39	110	8.1%
Middle Age (31-54)	383	701	79.6%
Old (55+)	59	189	12.3%

7. Integrated Strategic Action Plan

Synthesizing insights across workbook stages and filtered slicer scenarios yields three strategic directives:

1. Prioritize Short-Commute Suburban Corridors

Focus advertising and retail placement on residents living within 0–2 miles of urban/suburban workplaces. Emphasize commuting convenience, reduced fuel expenses, and health benefits.

2. Target Middle-Aged Single Professionals

Structure primary digital campaigns around single professionals aged 31–54 earning \$55k+. Highlight mid-to-high end models suited for daily transit and personal wellness.

3. Dynamic Slicer-Driven CRM Campaigns

Leverage regional and education slicer segments (e.g., Bachelors graduates in North America/Pacific) to deploy tailored promotions and financing options for price-sensitive groups.